DATA ANALYTICS MINI PROJECT

Name: Siddhant Indurkar

Roll Number:SA259

Project Title: Student Performance Analysis and Prediction

Problem Statement: This project analyzes student performance to understand how factors like gender, parental education, and preparation affect academic scores. The goal is to identify patterns that help improve student outcomes.

Dataset Description:

- Dataset Name: Student Performance Dataset
- Source: Kaggle
- Rows: 1000

Columns:

IMPORT LIBRARIES

Code:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

Output:

No output

Explanation:

Libraries imported successfully for data analysis and visualization.

DATA LOADING

Code:

```
df = pd.read_csv("StudentsPerformance.csv")
df.head()
```

Output:

```
   gender parental level of education ... reading score writing score
0  male   some high school           ... 69             38
1  female high school                ... 36             84
2  male   high school                ... 58             39
3  male   high school                ... 72             54
4  male   some college              ... 56             71
```

Explanation:

Dataset loaded successfully and first 5 rows displayed.

DATASET SHAPE

Code:

```
df.shape
```

Output:

```
(1000, 7)
```

Explanation:

Dataset contains 1000 rows and 7 columns.

DATA TYPES

Code:

```
df.dtypes
```

Output:

```
gender                          object
parental level of education     object
lunch                           object
test preparation course         object
math score                      int64
reading score                   int64
writing score                   int64
```

Explanation:

Dataset includes both categorical and numerical columns.

MISSING VALUES

Code:

```
df.isnull().sum()
```

Output:

```
All columns → 0
```

Explanation:

No missing values found in dataset.

DATA CLEANING

Code:

```
df.drop_duplicates(inplace=True)
df.shape
```

Output:

```
(1000, 7)
```

Explanation:

No duplicate rows found. Dataset is clean.

STATISTICAL ANALYSIS — MATH SCORE

Code:

```
df['math score'].mean()
df['math score'].median()
df['math score'].mode()
df['math score'].std()
df['math score'].var()
df['math score'].max() - df['math score'].min()
(df['math score'].max() + df['math score'].min()) / 2
```

Output:

```
Mean = 65.525
Median = 66.0
Mode = 76
Std Dev = 19.90
Variance = 395.85
Range = 69
Mid-range = 64.5
```

Explanation:

Math scores show moderate variation among students.

STATISTICAL ANALYSIS — READING SCORE

Code:

```
df['reading score'].mean()
df['reading score'].median()
df['reading score'].mode()
df['reading score'].std()
df['reading score'].var()
```

Output:

```
Mean = 68.182
Median = 69.0
Mode = 93
Std Dev = 18.96
Variance = 359.59
```

Explanation:

Reading scores are slightly higher than math scores.

VISUALIZATIONS

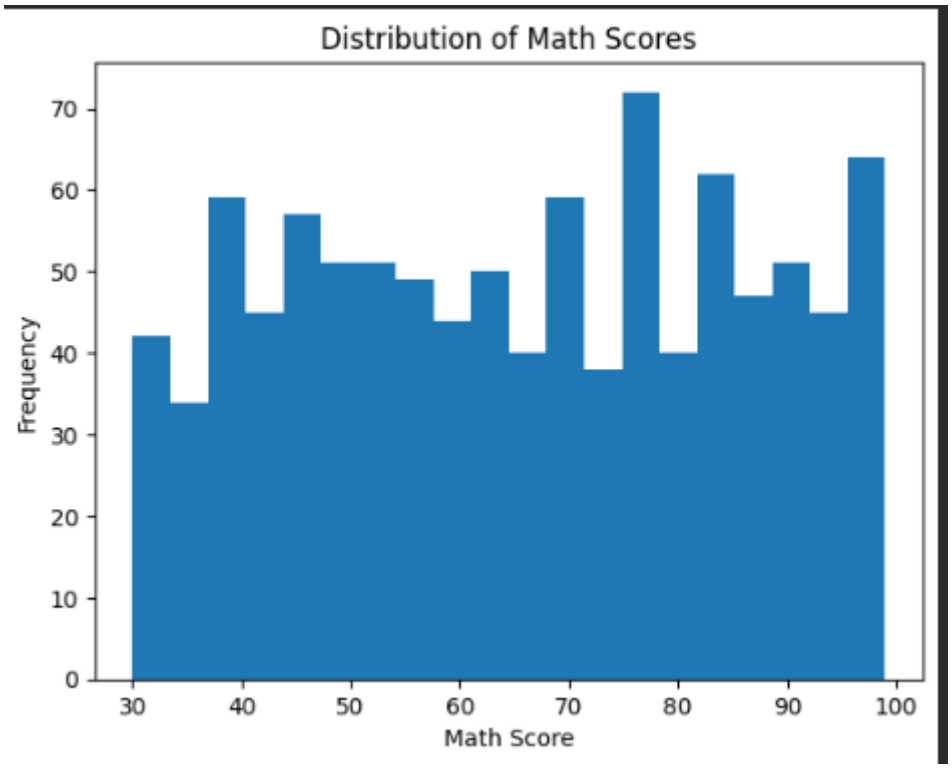
1. Histogram

Code:

```
plt.hist(df['math score'], bins=20)
plt.title("Distribution of Math Scores")
plt.xlabel("Math Score")
plt.ylabel("Frequency")
plt.show()
```

Output:

Histogram graph displayed.



Explanation:

Most students score between 40–90, showing moderate performance.

2. Gender Distribution

Code:

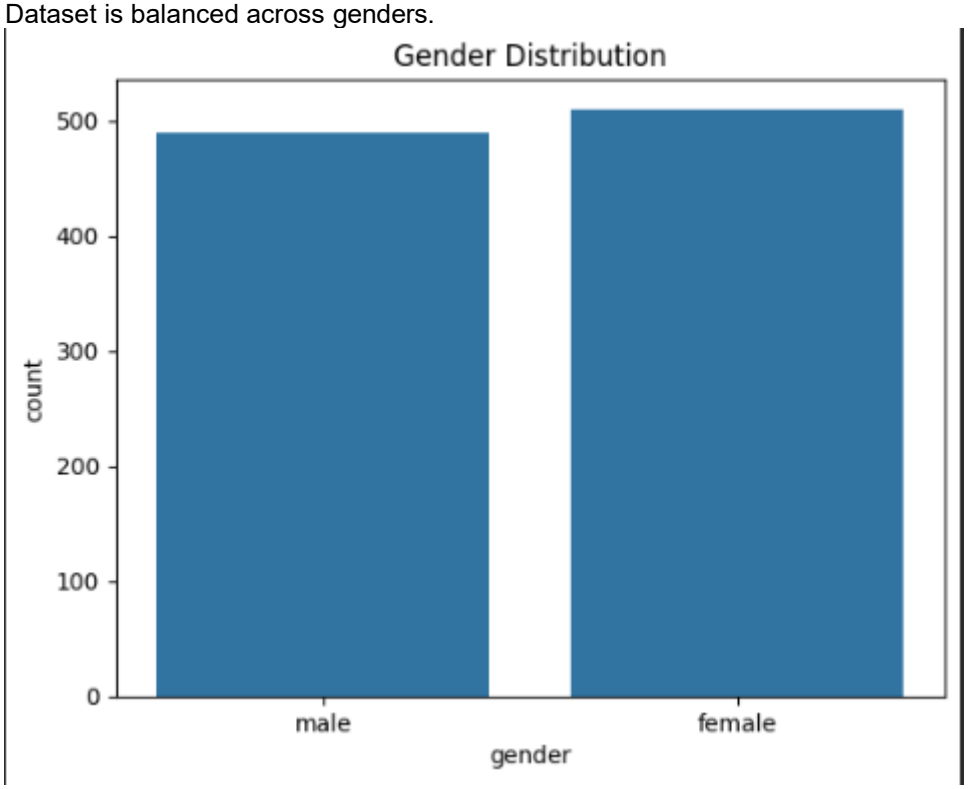
```
df['gender'].value_counts().plot(kind='bar')
plt.title("Gender Distribution")
plt.xlabel("Gender")
plt.ylabel("Count")
plt.show()
```

Output:

Bar chart displayed.

Explanation:

Dataset is balanced across genders.



3. Boxplot

Code:

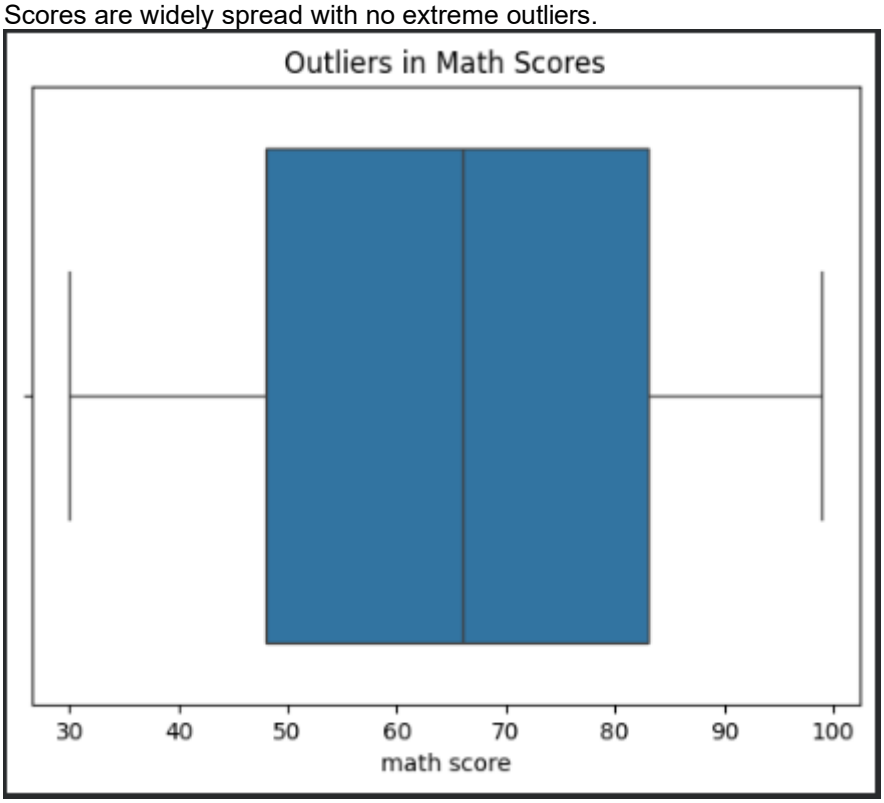
```
plt.boxplot(df['math score'])
plt.title("Math Score Spread and Outliers")
plt.show()
```

Output:

Boxplot displayed.

Explanation:

Scores are widely spread with no extreme outliers.



4. Heatmap

Code:

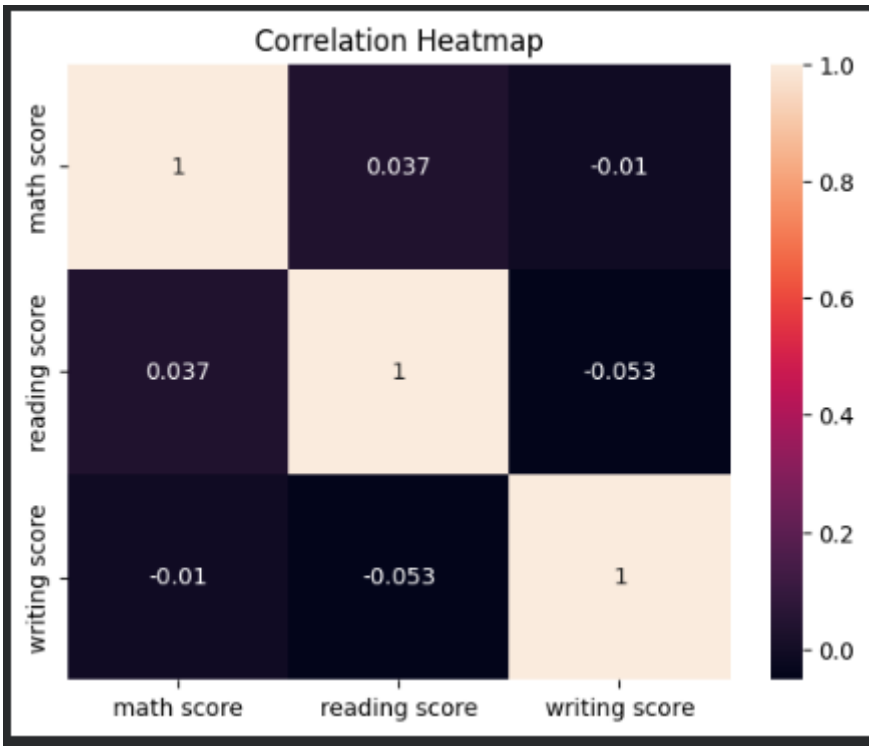
```
sns.heatmap(df[['math score','reading score','writing score']].corr(), annot=True)
plt.title("Correlation Heatmap")
plt.show()
```

Output:

Heatmap displayed.

Explanation:

Very weak correlation between subjects.



MODEL

Code:

```
from sklearn.linear_model import LinearRegression
model = LinearRegression()
model.fit(X_train, y_train)
model.score(X_test, y_test)
```

Output:

```
0.0016
```

Explanation:

Model shows very low accuracy, indicating weak prediction capability.

FINAL INSIGHT

The model shows very low accuracy, indicating that math performance cannot be predicted using only reading and writing scores. Other factors significantly influence performance.

Recommendations:

1. Focus on weak students identified via outliers
2. Use integrated learning for reading-writing improvement
3. Provide additional support for low-performing groups

GitHub link: https://github.com/rameshindurkar6521-max/Student-Performance-Analysis-and-Prediction